



DISPLAY REFERENCE

PowerGlove Vision Matrix Display Guide

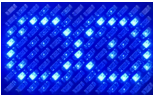
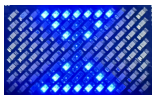
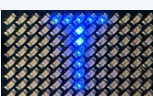
Recognize animations, mode letters, pairing, and startup feedback.

2026 EDITION / IAIN BENNETT / MIT LICENSE

The blue LED matrix on the **PowerGlove Vision Controller (Arduino UNO Q)** is PowerGlove Vision's status display. It tells you which mode is active and helps distinguish startup, practice, tracking, and pairing. It does not show a game's score or confirm that a game accepted a button.

Use the photographs and descriptions below to recognize the display and decide what to do next.

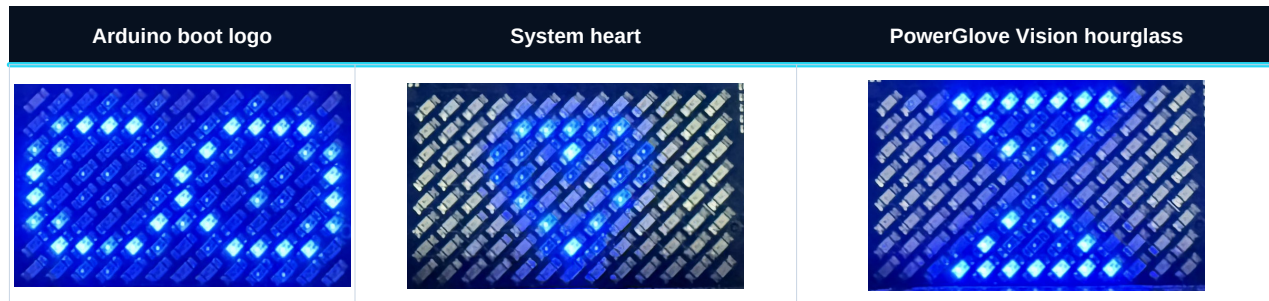
Recognize the display

What you see	See it	What it means	What to do
Arduino boot logo		The PowerGlove Vision Controller's system software is starting, before PowerGlove Vision controls the display.	Wait for the app's hourglass or normal display.
System heart animation		The board is progressing through system startup.	Wait for the app display.
Pulsing hourglass		PowerGlove Vision is starting.	Allow startup to finish. If it persists, check Dashboard.
Glove animation with moving cuff, curling fingers, and a spark		Gestures are off; the app is in its idle mode.	Open Glove Academy to practice, or choose a game profile on Dashboard.
A large scanning L		Play or Glove Academy lessons are active. L stands for local play or lessons.	Follow the game or practice moves shown in your browser; controller output is paused.
A large scanning T		Gesture tuning is active, including hand setup.	Follow the recording, preview, and save instructions in Glove Academy. Controller output is paused.
A steady A-I , BS , or GB		A game profile is selected, but a calibrated hand is not currently being reported as tracked.	Show your hand and check tracking/calibration on Dashboard.
A-I , BS , or GB gently changing brightness		The app reports a detected, calibrated hand for that profile.	Check Dashboard's controller status before playing. This pulse alone does not mean controls are enabled.
ID , characters, PN , and digits repeating	-	Secure pairing is showing the device identity and temporary PIN.	Follow the Connection page; read each group in order.
A flashing X		The app has requested an error display.	Read Dashboard's error message before deciding whether to reconnect the camera or restart.
A blank matrix		The display has been turned off, the app is stopping, or the board is still starting. It may also have lost power.	Use the browser and board power indicators to distinguish these cases. Blank does not prove shutdown is complete.

Startup: logo, hourglass, then your mode

A typical startup with **Gestures off** selected is:

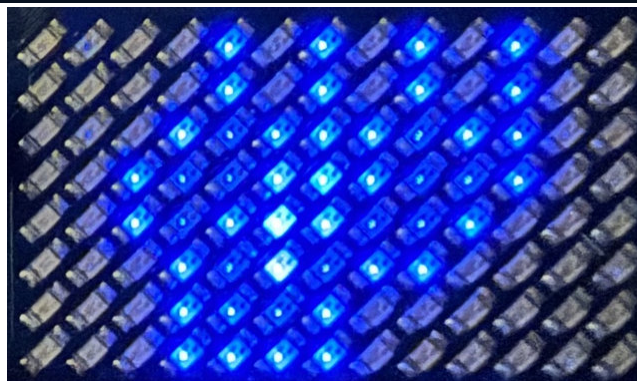
1. The board shows its Arduino boot logo and system heart animation.
2. The hourglass appears while PowerGlove Vision starts.
3. Once startup finishes, the glove animation appears when gestures are off.



With an active startup profile, the later display can instead be its profile letters. Opening Play or Glove Academy selects **L**; enabling tuning selects **T**. Wait for the camera view and startup message in your browser before practicing or playing.

The hourglass means startup is in progress. If it stays on the display, open Dashboard and check the startup or error message.

The idle glove show



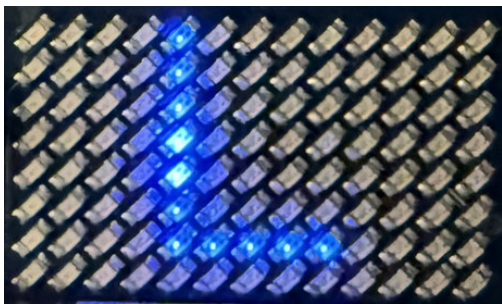
Idle glove: gestures are off; one frame of the repeating animation.

The glove curls, reopens, and glows while gestures are off. Open Glove Academy to practice, or choose a game profile on Dashboard when ready to play.

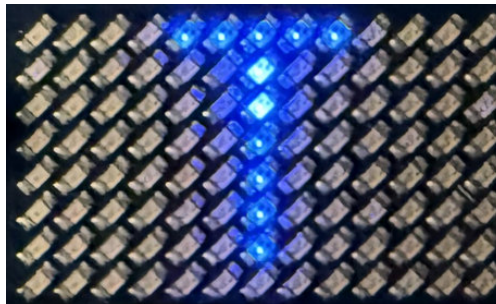
A flashing spark here does **not** mean you performed Glove Zap. This animation means gestures are off. It is different from selecting a game profile and merely pressing **Stop controller**: that keeps the camera/profile active and can leave profile letters visible.

Local play and Glove Academy: L and T

L: Local play or Glove Academy lessons



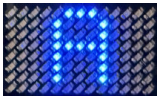
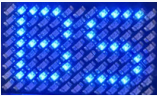
T: Gesture tuning



L means local Play or Glove Academy lessons are active. **T** means gesture tuning is active, including **Set up my hand**, individual adjustments, and previews. Follow the browser's instructions to practice or complete your recordings.

Both modes pause controller output. After practice, use Dashboard to check the selected profile and controller state; after tuning, explicitly start controller delivery when ready to play. A pairing display can temporarily cover either letter. Tracking details and pose failures remain in the browser rather than being spelled out on the matrix.

Game profile letters

Letters	See it	Selected profile
A through I		The corresponding reusable Program A-I profile; A is shown
BS		Bad Street Brawler
GB		Super Glove Ball

For example, **GB** is what you should expect with Super Glove Ball selected. It stays steady until a calibrated hand is being tracked, then gently pulses. The characters remain the same: the animation does not identify individual finger curls, movements, or button presses.

The important distinction is **tracking versus delivery**. A pulsing **GB** can appear while controller output is stopped. Confirm **Start controller** has been used and Dashboard shows output enabled, then confirm the action in the game. The matrix does not acknowledge RetroPie receipt or the game's response.

A generic **PG** ready symbol or a small pulsing tracking symbol can appear when no recognized profile identifier is available. These are fallback displays; normal named game profiles use their letters. They do not represent extra games or new gesture commands. If you expected **GB** or **BS**, check the selected profile on Dashboard.

Pairing: ID and PN

During secure pairing, the small display presents the information in pieces:

1. **ID** announces the device certificate identity.
2. Seven hexadecimal characters follow in groups of two, with the last character shown alone.
3. **PN** announces the temporary six-digit PIN.
4. Three pairs of digits follow, preserving leading zeroes.
5. The sequence repeats so you can read it again.

Read each pair from left to right and join the groups in the order shown. Do not mistake **PN** for a game program or use the certificate identity as the PIN.

Follow the Connection page to compare the identifier and enter the PIN. Pairing information temporarily replaces the usual display. If it expires before you finish, prepare a new pairing attempt. Check the page for confirmation that pairing succeeded.

For public photographs, use a clearly marked demonstration sequence or obscure live pairing digits. Never publish an active PIN or connection credentials. See [secure pairing](#) for the full procedure.

Errors, pauses, and a blank display



A blinking **X** means the app needs attention. Read Dashboard's error message for the cause and the next step. Check Dashboard whenever something is not working, even if the matrix still shows a normal mode.

A blank display alone does not prove the board is safe to unplug. Use the normal Shutdown procedure and its completion guidance. If the board should be running, check its power, wait for boot to finish, and try Dashboard. If the web app works but the display stays blank, record what appeared immediately before it went blank and whether restarting the app changes it.